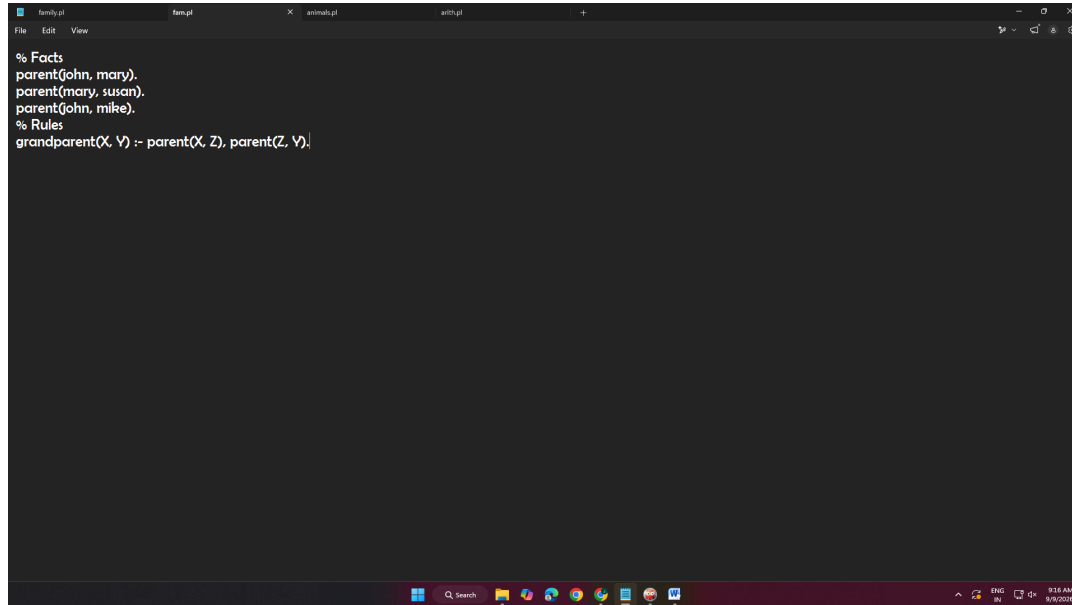


## Exercise :6 Programs in Prolog

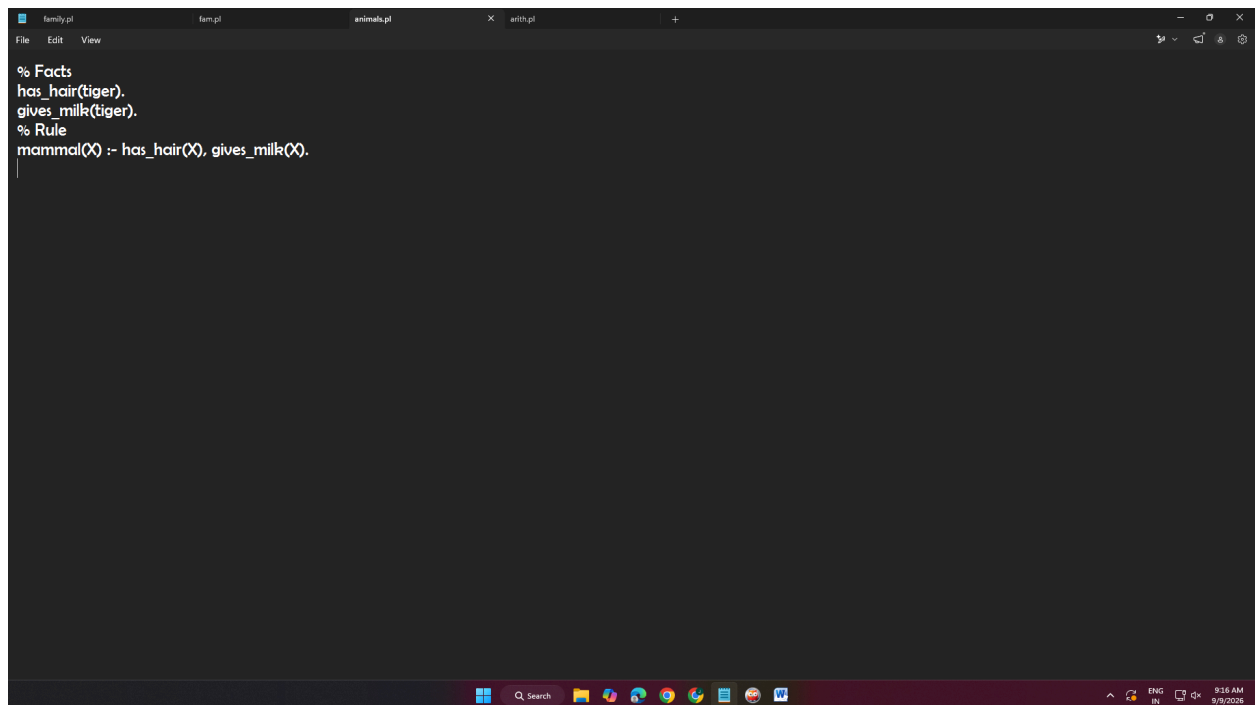
### 1.FAMILY RELATIONSHIP:

A screenshot of a Prolog IDE window with three tabs: 'family.pl', 'animals.pl', and 'arith.pl'. The 'family.pl' tab is active and contains the following Prolog code:

```
% Facts
parent(john, mary).
parent(mary, susan).
parent(john, mike).
% Rules
grandparent(X, Y) :- parent(X, Z), parent(Z, Y).
```

The IDE has a menu bar with 'File', 'Edit', and 'View'. The Windows taskbar at the bottom shows the Start button, a search bar, and several application icons. The system tray on the right indicates the language is 'ENG IN' and the time is '9:16 AM 9/9/2026'.

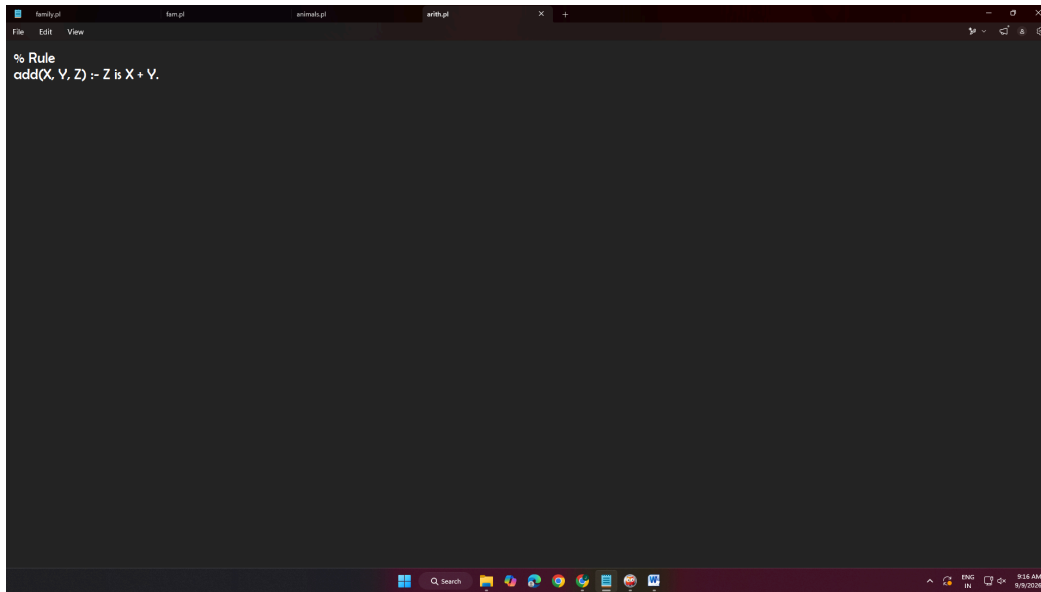
### 2.ANIMAL CLASSIFICATION:

A screenshot of a Prolog IDE window with three tabs: 'family.pl', 'animals.pl', and 'arith.pl'. The 'animals.pl' tab is active and contains the following Prolog code:

```
% Facts
has_hair(tiger).
gives_milk(tiger).
% Rule
mammal(X) :- has_hair(X), gives_milk(X).
```

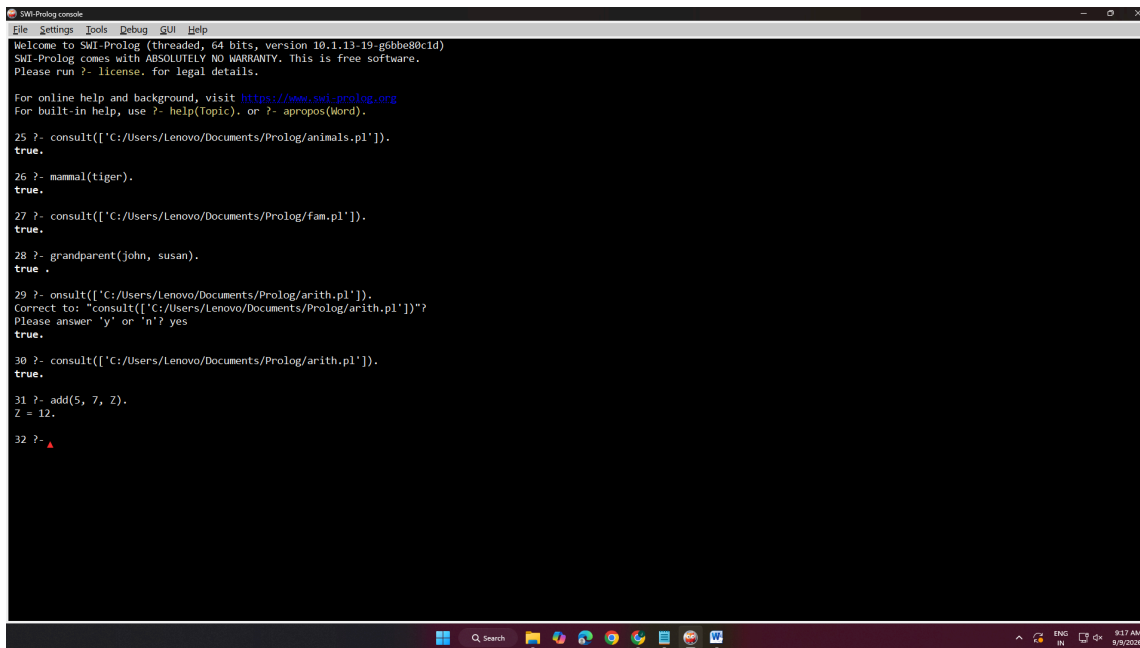
The IDE has a menu bar with 'File', 'Edit', and 'View'. The Windows taskbar at the bottom shows the Start button, a search bar, and several application icons. The system tray on the right indicates the language is 'ENG IN' and the time is '9:16 AM 9/9/2026'.

### 3.SIMPLE ARITHMETIC KNOWLEDGE BASE:



The screenshot shows a Prolog editor window with a dark theme. The title bar indicates three files are open: 'fam.pl', 'animals.pl', and 'arith.pl'. The 'arith.pl' file is the active one. The code in the editor is a single line defining a rule: `% Rule`  
`add(X, Y, Z) :- Z is X + Y.`

### OUTPUT:



The screenshot shows the SWI-Prolog console window. It displays the following text:

```
SWI-Prolog console
File Settings Tools Debug GUI Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.1.13-19-g6bbe80c1d)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

25 ?- consult(['C:/Users/Lenovo/Documents/Prolog/animals.pl']).
true.

26 ?- mammal(tiger).
true.

27 ?- consult(['C:/Users/Lenovo/Documents/Prolog/fam.pl']).
true.

28 ?- grandparent(john, susan).
true.

29 ?- consult(['C:/Users/Lenovo/Documents/Prolog/arith.pl']).
Correct to: "consult(['C:/Users/Lenovo/Documents/Prolog/arith.pl'])"?
Please answer 'y' or 'n'? yes
true.

30 ?- consult(['C:/Users/Lenovo/Documents/Prolog/arith.pl']).
true.

31 ?- add(5, 7, Z).
Z = 12.

32 ?- ▲
```